

HTL Leoben	HTL Leoben Higher Federal Technical College of Industrial Engineering – Information technology and smart production	Diploma Exam
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DIPLOMA THESIS Documentation

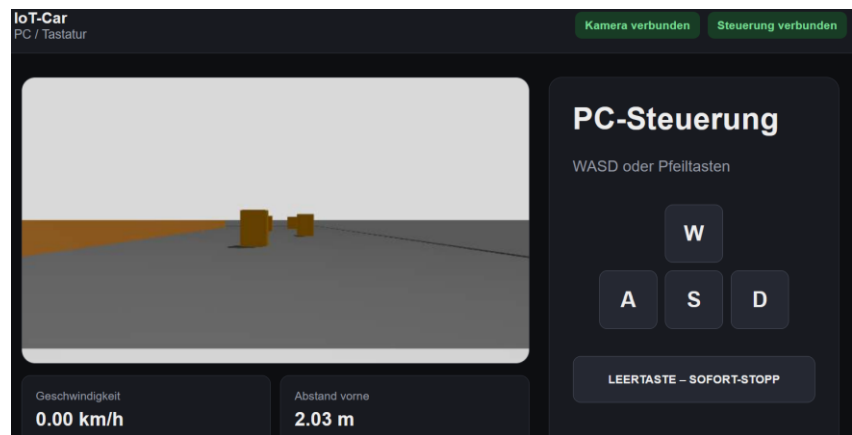
Author(s)	Chloe Pripfl
Form, academic year	5BHWIN, 2025/26
Topic	Creation of the ROS2 simulation
Co-operation partners	HTL Leoben

Assignment of tasks	Development of a functional simulation of the HTL Leoben IoT car. The vehicle is to be recreated virtually in Gazebo Sim and controlled using ROS 2. In addition, a front camera, a distance sensor and speed measurement are to be integrated. The vehicle is to be operated through a web-based interface for desktop computers and smartphones. Finally, the complete application is to be provided in a Docker environment.
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Realisation	Completion in September of 2026
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Results	The planned result is a functional and reproducible IoT car simulation including vehicle control, sensors and a web interface. The vehicle should be able to drive forwards and backwards and be controlled using Ackermann steering. Camera images, speed and distance data should be displayed in real time in the web interface. In addition, the simulation should be containerized so that it can be started and further developed easily on compatible systems.
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Illustrative graph, photo
 (incl. explanation)



A screenshot of the control panel on a computer

Participation in competitions
 Awards

Accessibility of the
 Diploma Thesis

Approval
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Examiner

Head of College